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> #Date: December 3, 2024
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> #This code reads in the MPG data set, creates the cars data frame, and then
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> #uses the cars data frame to conduct a hypothesis test to evaluate if there is a
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```
> #significant difference between the average gas mileage of cars manufactured
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```
> #in the US and cars manufactured in Japan. This test assumes unequal variances.
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```
> #Read in the MPG data set and creates the cars data frame.
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```
> cars<-read.csv("mpg.csv")
```

```
> cars
```

```
Country MPG
```

```
1  US 18
```

```
2  US 15
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```
3  US 18
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```
4  US 16
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5  US 17
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6  US 15
```

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7  US 14
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8  US 14
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9  US 14
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10 US 15
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11 US 15
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12 US 14
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13 US 15
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14 US 14
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15 US 22
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16 US 18
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17 US 21
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18 US 21
19 US 10
20 US 10
21 US 11
22 US 9
23 US 28
24 US 25
25 US 19
26 US 16
27 US 17
28 US 19
29 JP 24
30 JP 27
31 JP 27
32 JP 25
33 JP 31
34 JP 35
35 JP 24
36 JP 19
37 JP 28
38 JP 23
39 JP 27
40 JP 20
41 JP 22
42 JP 18
43 JP 20
44 JP 31
45 JP 32

46 JP 31

47 JP 32

48 JP 24

49 JP 26

50 JP 29

51 JP 24

52 JP 24

53 JP 33

54 JP 33

55 JP 32

56 JP 28

>

> #Using cars data frame to conduct a hypothesis test to evaluate if there is a

> #significant difference between the average gas mileage of cars manufactured

> #in the US and cars manufactured in Japan. This test assumes unequal variances.

> with(cars, t.test(MPG~Country, var.equal=FALSE, alternative="two.sided", mu=0))

Welch Two Sample t-test

data: MPG by Country

t = 8.533, df = 53.657, p-value = 1.444e-11

alternative hypothesis: true difference in means between group JP and group US is not equal to 0

95 percent confidence interval:

7.895995 12.746863

sample estimates:

mean in group JP mean in group US

26.75000 16.42857

